Day - 2

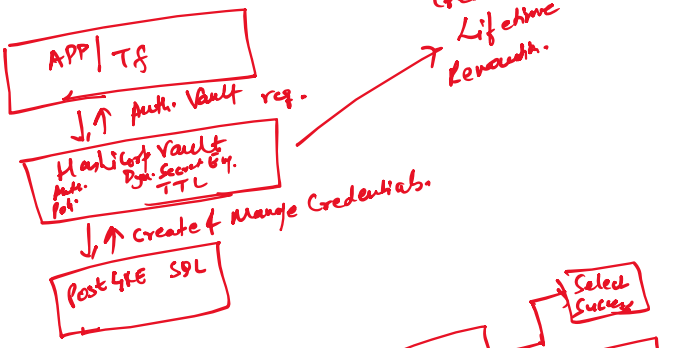
03 November 2026 09:40

1. Dynamic Secret engine.
2. Policies & Access Control
3. Auth Methods Deep Dive.
4. Lease Renewal & Revocation.
5. Response wrapping & cubbyhole.

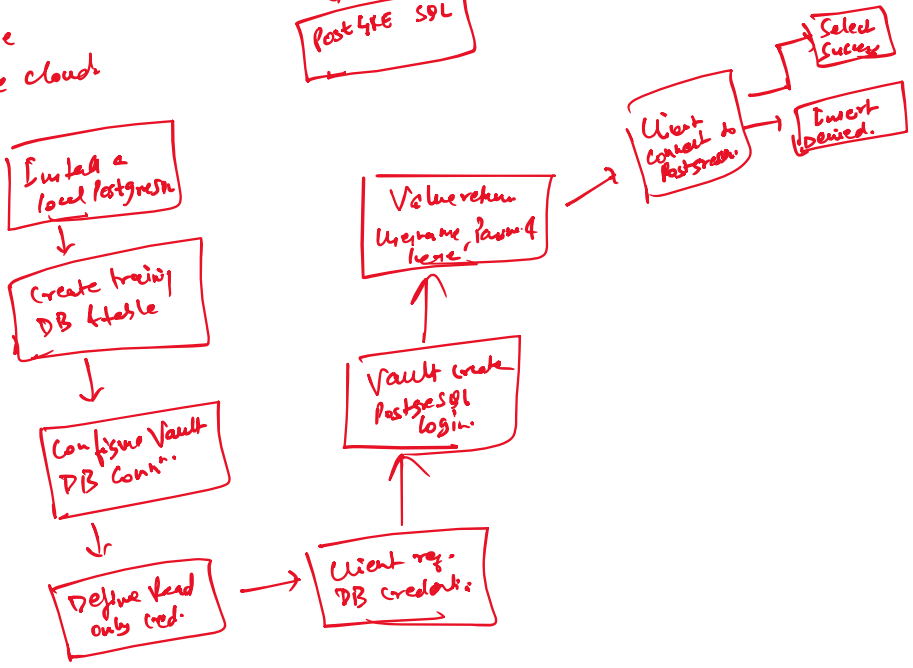
1. Dynamic Secret Engine:- Generate temp. credentials on demand.

Unique





1. AWS
2. DB
3. PKI
4. Azure
5. Google cloud

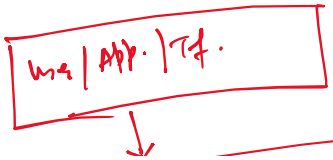


2. Policies & Access Control - ACL System, Least Privilege:-

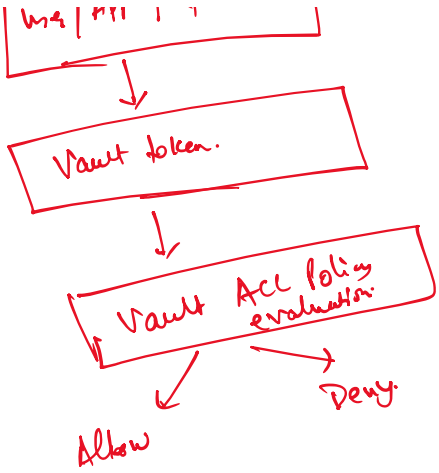
1. What Authenticated

Authentication → Who?

Authorization → What !



HCL



HCL
↓
HashiCorp Config. Lang.

path "< Vault-api-path >" {
Capabilities = ["read", "update"]
}

ACL → create, read, update, delete, list, Patch, sudo, deny.
↓ Modify
↓ Partial update
↓ refine.

ACL Path:- foundation |

foundation | dev | app.

" " * |

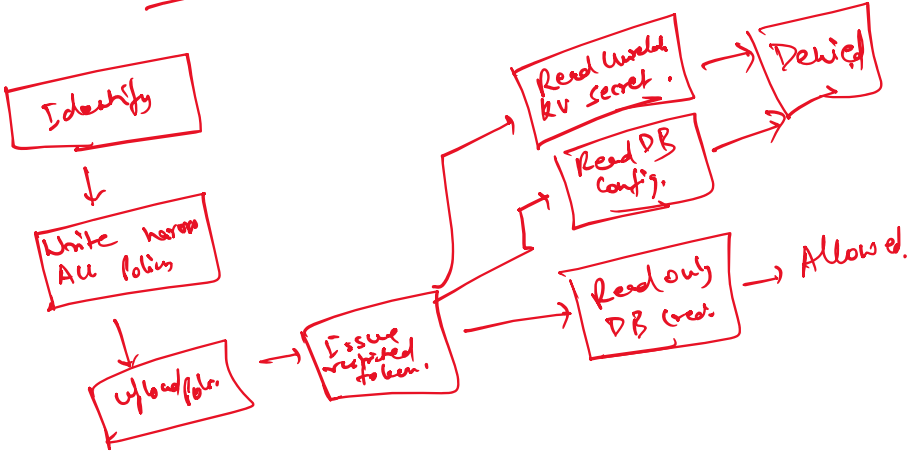
foundation | data | + | app

CLI

KV V2 path distinction — vault kv put foundation | dev | app

foundation | metadata | dev | app

PoLP → Principle of least Privilege.



(3) Authentication Method Deep Dive! —

(1) App Role.

Machine to Machine. & Workload

② k8s
③ JWT | OIDC

Authentication.

AppRole

RoleID + SecretID

VM, APP, Jenkin.

No extend broker

k8s

Service Account.

Pod

k8s API

JWT | OIDC

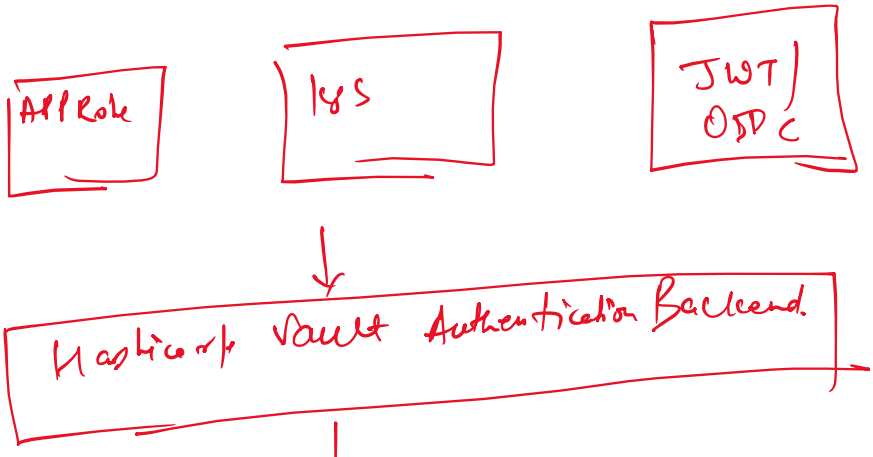
Signed JWT

OIDC identity

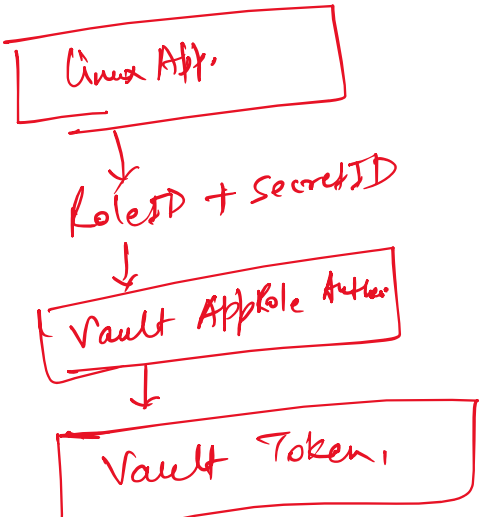
CI/CD,

SSO

external provider.



AppRole :-



Vault Token.

K8S :-

EKS | AKS | GKE

↓

Service Account

↓

Vault K8S Auth

↓

Vault token → ACL → App-Secret

JWT | OIDC :-

JSON Web Token.

Open ID Connect

JWT →

Header | Payload | Signature.

OIDC :- Open ID Connect.

OAuth 2.0

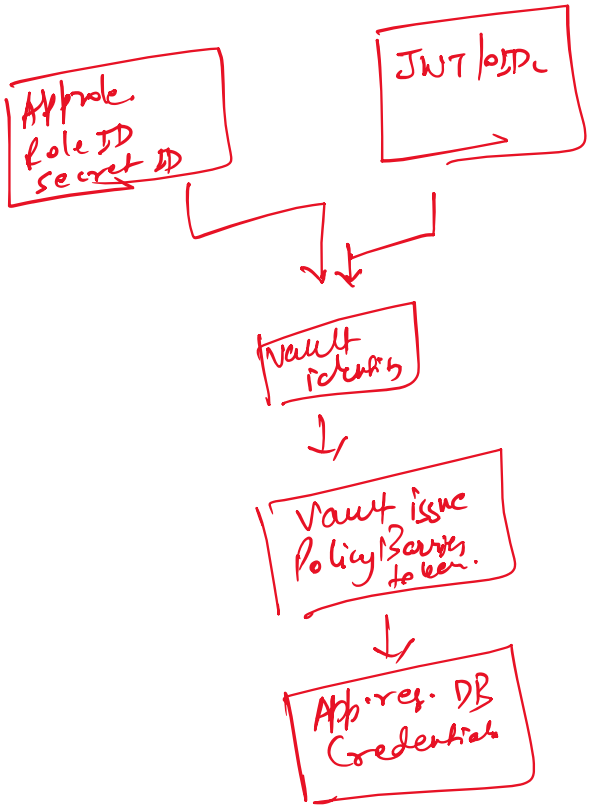
HCP Terraform Workspace.

↓

HCP Terraform Signed JWT

↓

Vault JWT Authentication. → Vault Token + Policies.



token lifetime.

max

```
vault write auth/approle/role/training-app token_policies="db-reader" token_ttl="10m" token_max_ttl="30m" secret_id_ttl="10m"
secret_id_num_uses=1
```

↳ Once.

4. Lease, Renewal & Revocations

1. How long valid
2. How to extended
3. How access can be terminated?

1. Lease + Time Bound.

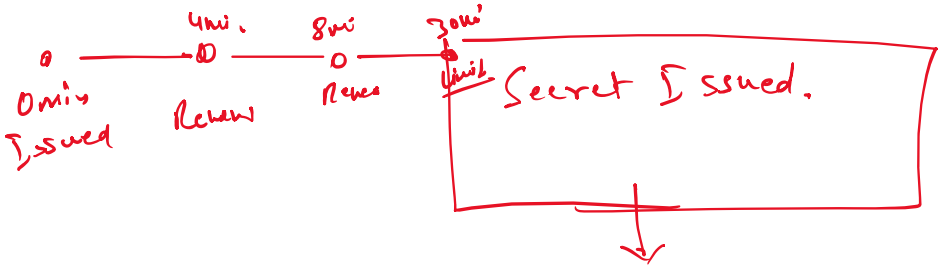
```
ubuntu@ip-172-31-23-11:~$ VAULT_TOKEN="$APPROLE_TOKEN" vault read database/creds/readonly
Key                Value
---                -----
lease_id           database/creds/readonly/oLwhySUL3I53Pzw76krXM7Fh
lease_duration     5m
lease_renewable    true
```

UID

UID

TTL
true/false.

```
---             -----
lease_id        database/creds/readonly/oLwhySUL3I53Pzw76krXM7Fh
lease_duration  5m
lease_renewable true
password        yZukgDQ-nvcJWOzkw1xT
username        v-approle-readonly-f5eQUu4ZVZK33Tow3dtn-1793695772
ubuntu@ip-172-31-23-11:~$ VAULT_TOKEN="$APPROLE_TOKEN" vault read database/config/local-
postgres
```



Active lease - TTL 5min.

Renew
extend

expire | Revoke
Credential Revocation

(2) TTL (Time to live):- How long?

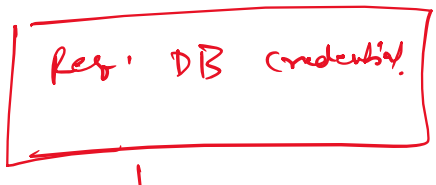
← default_ttl = 5m
max_ttl = 30m. →

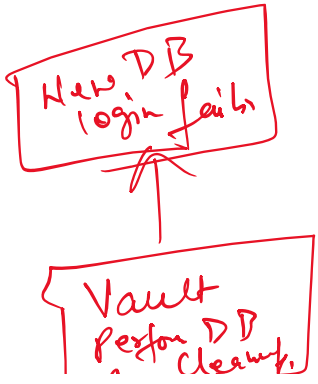
(3) Renewal - Extending a lease:-

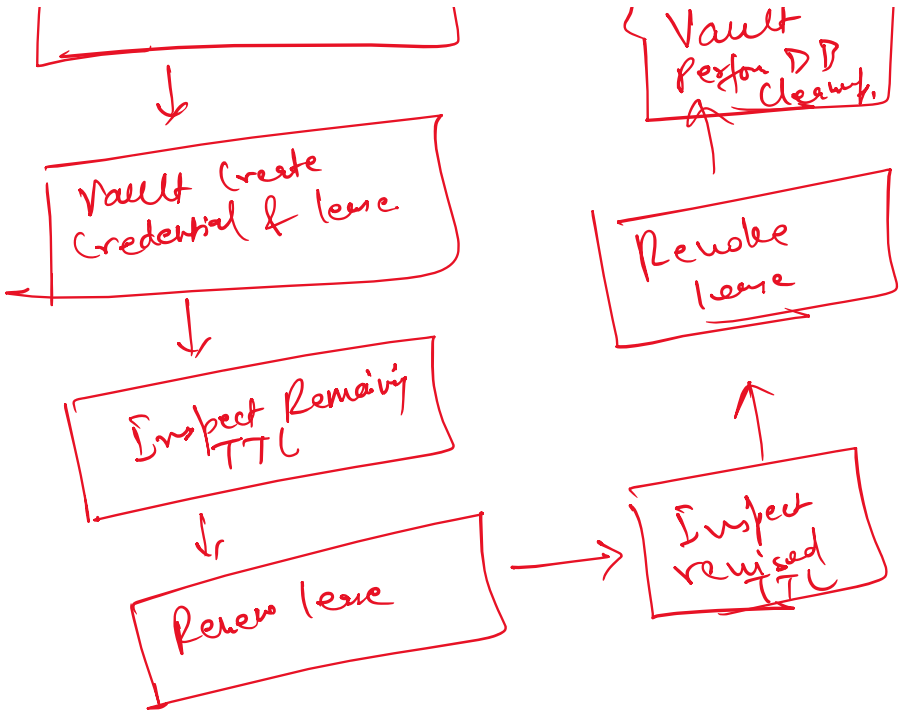
(4) Revocation - Terminating credentials B/f expiry

(5) Token Renewal vs Secret lease renewal.

Lab →

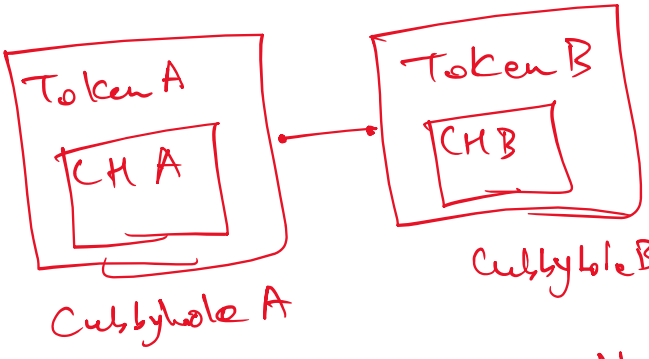






(4) Response Wrapping & Cubbyhole :-

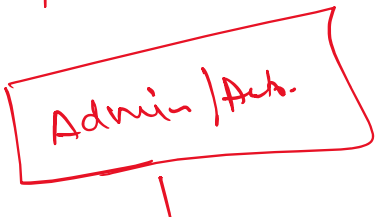
(1) Cubbyhole → Private, token-specific storage.



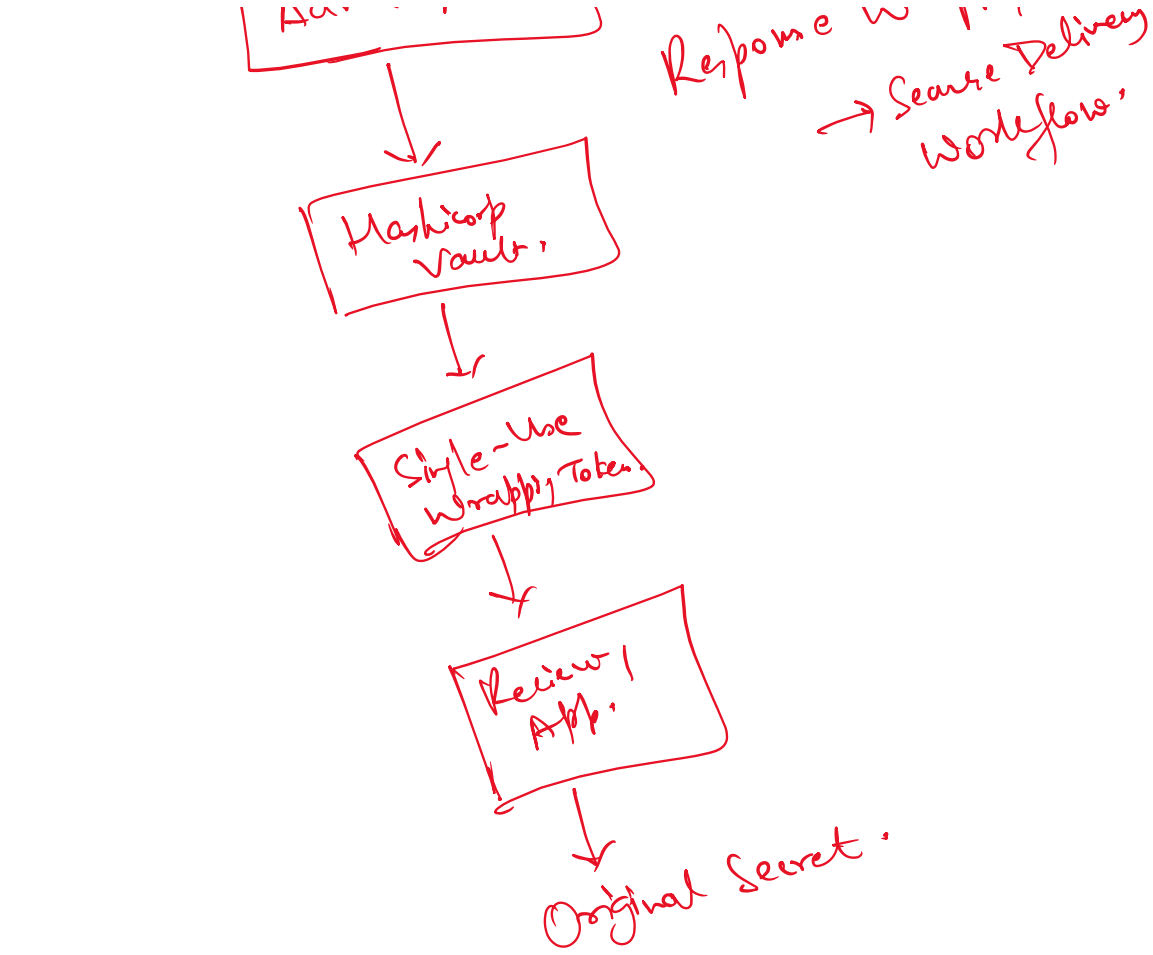
Token A → Cubbyhole B

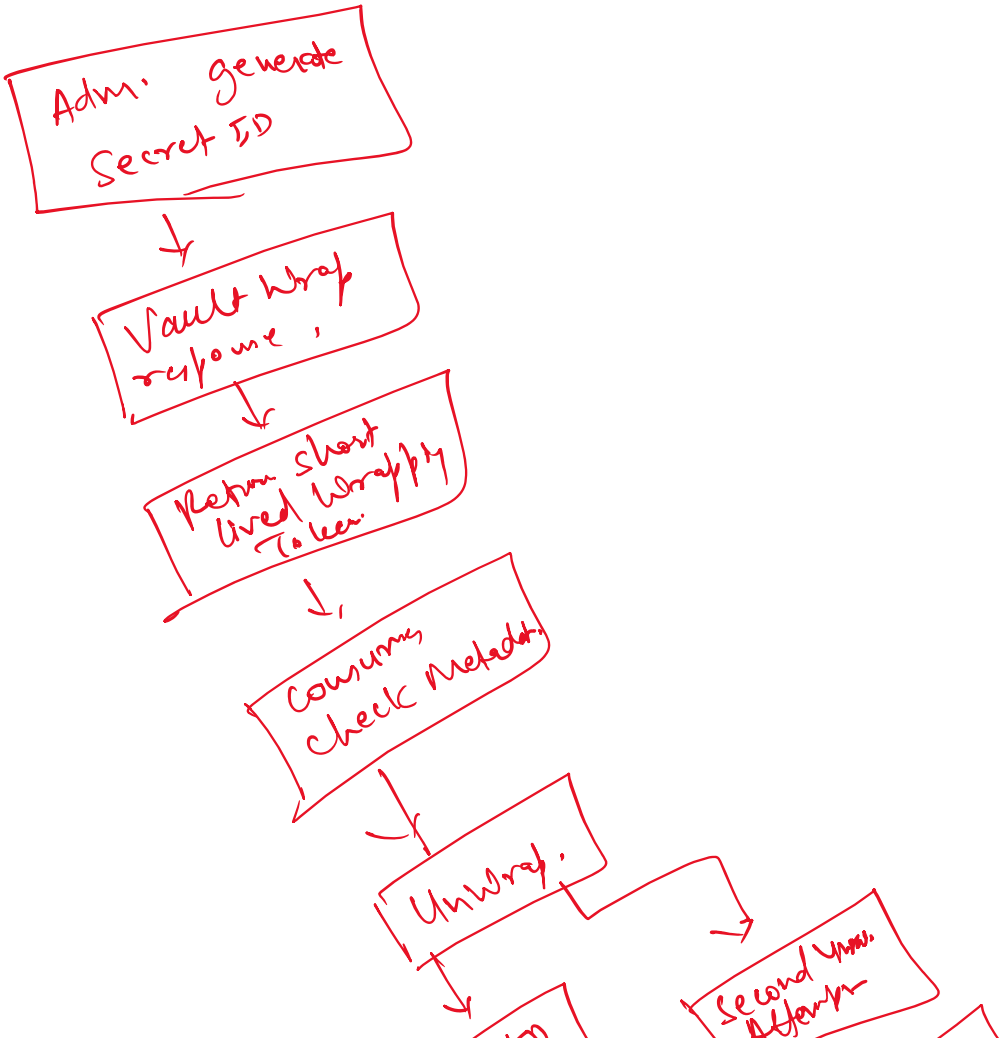
(2) Response Wrapping :-

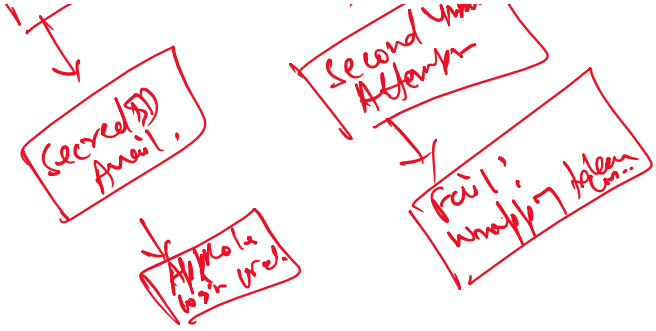
Temp. Hide Vault API



Cubbyhole → storage Mechanism.
Response Wrapping
- Secure Delivery


Response w
→ Secure Delivery
Workflow,
Hashicorp
Vault,
Style-use
Wrapping Token,
Review
App,
Original Secret.


Adm. generate
Secret ID
Vault wrap
response,
Return short
lived Wrapping
Token
Consumer
check Metadata
UnWrap,
Second Level
Authen



Day-3

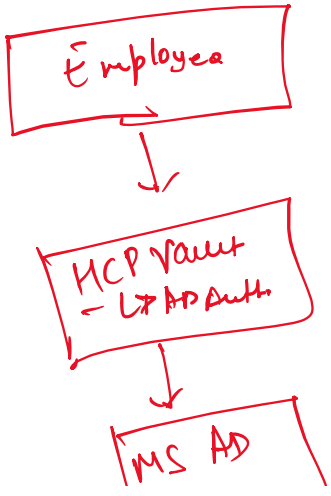
1. PKI Secret Engine.
2. Transit secret Engine.
3. Auto-Unseal.
4. Vault Agent & Vault Agent Templates
5. HA, Replication & DR
6. Monitoring, Audit Device & Best Practices.

HCP Tf → Signed JWT → Vault JWT Auth

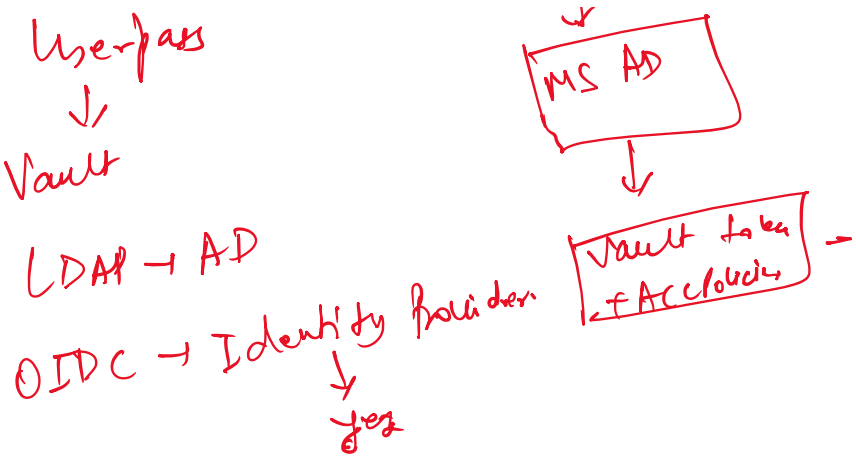
↓

KV secret ← Vault Token

LDAP →



Userpass


User-pass
Vault
LDAP → AD
OIDC → Identity Provider
yes
MS AD
Vault token
+ Accessors